DATA ANALYTICS REPORT

What is really driving customer experience on the Olist marketplace

An evidence-based analysis of 99,441 Brazilian e-commerce orders, September 2016 – October 2018

PREPARED FOR	DATASET	EVIDENCE BASE	COMPANION
Olist leadership — operations & CX	**Brazilian E-Commerce Public Dataset (Olist)**	**9 tables · 1.55M rows · 96,470 delivered & reviewed orders**	**olist_marketplace_analysis.ipynb (64 cells, runs clean)**

R$13.6M	**8.1×**	**4.07**	**15.1%**	**8.1%**
product revenue over 25 months	order growth Jan 2017 → Aug 2018	mean review score 57% give 5 stars	of orders rated 1–2 stars	of deliveries miss the promised date

THE FINDING IN ONE PARAGRAPH

Olist scaled order volume eightfold without degrading customer satisfaction — mean review score shows no significant downward trend across the period and ends higher than it started. But satisfaction swings sharply month to month, and those swings track the **late-delivery rate** ($\rho = -0.48$, $p = 0.031$), not order volume ($\rho = -0.08$, $p = 0.75$). At order level the mechanism is unambiguous: a delivery that misses its promised date is rated 1–2 stars **54.6%** of the time versus **9.5%** when it arrives on time — a 5.8× risk ratio holding in all 5 regions and all 32 testable categories.

Late orders are just 8.1% of deliveries yet generate 33.7% of all bad reviews; delivery failure as a whole explains **48.5%** of them. Geography, product category and freight cost look like separate problems but largely dissolve once delivery timing is held constant. The remaining **51.5%** of bad reviews come from orders that arrived on time and fast — and the customers' own written comments say those are about the **product**, not the parcel.

CONTENTS

SECTION 1

Problem Understanding

What leadership asked, what the data can actually answer, and what would count as a good answer

The business situation

Olist is a marketplace integrator: small and medium Brazilian merchants sign one contract and sell through the major marketplaces without negotiating each individually. When an order is placed, the responsible seller is notified, fulfils it, and ships through Olist's logistics partners. Once the order is delivered — or once the estimated delivery date has simply passed — the customer receives a satisfaction survey and can leave a 1–5 star review with optional written comments.

That last detail matters more than it first appears, and it shapes the whole analysis. The survey fires on the **promised** date, not only on delivery. Customers whose orders never arrived are therefore invited to review them, and they do. The promised date is not merely an internal planning number; it is the moment Olist itself asks the customer to render judgement.

Leadership holds roughly two years of history across orders, items, payments, products, sellers, customers, reviews and geolocation, but no consolidated analysis has connected these dimensions. Review scores, delivery timing, seller performance and payment behaviour all vary considerably across the period, across categories and across Brazil's 27 states — and nobody has established which of that variation is cause, which is consequence, and which is coincidence.

The question behind the questions

The brief poses six core questions. Underneath them sits a single decision problem: **where should a scaling marketplace spend its next unit of operational effort?** That framing drove three choices in this analysis.

- **Rank factors by recoverable damage, not by strength.** A factor can triple the risk of a bad review and still be unimportant if it is rare. Every candidate driver is therefore scored on **population attributable fraction** — how much of the platform's total dissatisfaction would disappear with it — which combines effect size with prevalence.
- **Separate drivers from passengers.** Remote regions, long distances and high freight ratios all correlate with bad reviews. If they matter only because they make lateness more likely, they are symptoms and should not receive their own programme. Every factor is therefore re-tested **within on-time deliveries only**.
- **Size the residual honestly.** Once delivery is accounted for, whatever dissatisfaction remains must be named and measured rather than waved away, because it points at a different owner inside the business.

What the data can and cannot support

This is observational data with no experiment in it, so no claim of proven causation is available at any price. What *is* available is a strong evidential case: a large effect, a clean dose–response, stability across every stratification tested, a plausible operating mechanism, and — unusually — independent corroboration from a second, unrelated signal in the written review text. Where that case is made, this report says so. Where the evidence supports only association, the wording stays at association.

TWO DATA TRAPS THAT WOULD SILENTLY CORRUPT THE ANSWER

Counting customers with the wrong key. `customer_id` is unique per order, not per person; `customer_unique_id` identifies the human. Using the former inflates the customer base by 3.5% and makes repeat-purchase analysis impossible. All customer counts here use the latter.

Joining tables at the wrong grain. The geolocation table holds ~1.0M coordinate readings for only 19,015 zip prefixes — a median of 29 readings each. Joined raw, it multiplies every order row and inflates revenue roughly fiftyfold. It is aggregated to one median coordinate per prefix before use. Similarly, 555 orders carry more than one review row; these are collapsed to one review per order before any join.

SECTION 2

Analytical Approach

From nine raw tables to a ranked, deconfounded set of drivers

STEP 1
Audit
Verify every stated key, foreign key and grain before trusting any join.

STEP 2
Clean
16 logged decisions, each with row counts and a stated reason. Nothing dropped silently.

STEP 3
Model
One order-level master table, 82 columns, exactly 99,441 rows — asserted, not assumed.

STEP 4
Test
Six core questions, each with effect sizes and confidence intervals, not just p-values.

STEP 5
Deconfound
Re-run every driver inside on-time deliveries to separate causes from symptoms.

STEP 6
Corroborate
Check the conclusion against the Portuguese review text — an independent signal.

Data preparation decisions that shape the results

Six decisions materially affect what the numbers say, so they are stated rather than buried. Each was logged in the notebook with the number of rows affected.

Table 1. Cleaning and modelling decisions with material consequences

Decision	Choice made	Why it matters
Non-delivered orders	Kept in volume and revenue; excluded from delivery-timing analysis	2,965 orders have no delivery date by construction. Imputing would fabricate deliveries; deleting would understate volume.
Duplicate reviews	Collapsed to one per order, keeping the latest answer (559 rows removed)	Otherwise those orders are double-counted in every downstream join.
Geolocation	Median latitude/longitude per zip prefix, inside Brazil's bounding box	Prevents a ~50× row explosion; the median resists mis-recorded GPS points that a mean would follow.
Missing category	Labelled `unknown`, rows kept (610 products)	They carry real revenue and real reviews — and turn out to be the worst-rated group on the platform.
Order-level attribution	Most expensive line item defines the order's category and seller	Only 9.9% of orders are multi-item; alternative rules were checked and change no conclusion.
Trend window	September–October 2018 excluded from all time series	The export stops mid-October; those months are truncated and would read as a demand collapse.

The statistical method, and why it is built this way

With ~96,000 observations, almost any comparison returns a significant p-value. Significance is therefore treated as a hygiene check, and every claim is carried by a magnitude instead.

Table 2. Statistical toolkit and the specific failure each choice guards against

Technique	Applied to	Guards against
Cliff's delta, Cramér's V, η^2	Every group comparison	Mistaking a significant-but-trivial difference for an important one.
Wilson score intervals	All proportions (1–2★ rates)	Over-reading small segments; Wilson stays valid at extreme rates and small n.
Rank-based tests (Spearman, Mann–Whitney, Kruskal–Wallis)	All continuous variables	Order value, freight and delay are heavily skewed; parametric tests would mislead.
Benjamini–Hochberg FDR	52 categories, 795 sellers	Testing 52 categories at α=0.05 yields ~3 false "weak categories" by chance alone.
Empirical-Bayes shrinkage	Seller scores	A seller with three 5-star reviews outranking a seller with 500 good ones.
Population attributable fraction	Driver ranking	Prioritising a rare-but-severe factor over a common one with more total damage.
Stratified 2×2 and within-stratum re-testing	Promise vs. wait; all drivers	Two correlated variables being credited with each other's effect.
Binomial test vs. platform rate	Seller lateness flags	Flagging a 20-order seller on the same evidence as a 500-order seller.

THE ANALYTICAL SPINE: THREE QUESTIONS ASKED IN ORDER

1. Does it matter? — effect size and attributable fraction for twelve candidate factors, all measured on the same population so they are directly comparable. **2. Does it survive?** — every factor re-tested inside on-time deliveries only; whatever keeps its risk ratio has an effect of its own, whatever collapses was a proxy for lateness. **3. Does anything else agree?** — the conclusion is checked against 38,764 written Portuguese comments, a signal with no timestamps in it at all.

Populations used

Every figure in this report names the population it describes. Four are used: **all orders** (99,441) for volume and revenue; **delivered with a real delivery timestamp** (96,470) for timing; **delivered and reviewed** (96,470) for every satisfaction claim; and **complete months only** (through August 2018) for trends. In this dataset essentially every order carries a review record, so the reviewed and delivered populations coincide.

SECTION 3

Key Insights

Eight findings, each with the evidence that carries it

01 The platform scaled 8× without degrading — growth is not the problem

Orders grew from 800 in January 2017 to 6,511 in August 2018 while average order value stayed flat near R$139, so revenue is a pure volume story. Mean review score shows *no* significant time trend over those 20 months and ends higher than it began. The intuitive "growth outran quality" narrative is not supported, and saying so matters: it redirects attention from scale to execution.

> **Evidence:** volume vs time ρ = +0.82 (p < 0.001) · score vs time ρ = +0.34 (p = 0.139, n.s.), linear slope +0.004 stars/month (p = 0.47) · Jan 2017 4.05 → Aug 2018 4.25

02 Satisfaction tracks delivery reliability, month by month

What the score does instead of trending is swing — and the swings are not random. Monthly mean score correlates with that month's late-delivery rate at $\rho = -0.48$, but with order volume at only −0.08. The three worst months for ratings are the three worst months for delivery: March 2018 (3.73 stars, 21.4% late), February 2018 (3.81, 16.0%) and November 2017 (3.89, 14.3%, Black Friday). Each is followed by recovery as the backlog clears. Volume growth does not hurt customers; volume growth that outruns the logistics operation does.

> **Evidence:** late rate vs score $\rho = -0.48$ ($p = 0.031$) · volume vs score $\rho = -0.08$ ($p = 0.75$) · best month Jun 2018: 4.27 stars at 1.4% late

03 Missing the promised date is the single largest driver of dissatisfaction

At order level the effect is large, monotonic and universal. Late orders are rated 1–2 stars 54.6% of the time against 9.5% for on-time orders, with mean score falling from 4.28 to 2.54. There is no threshold artefact: 0–3 days late already lifts the 1–2★ rate to 20%, and beyond 30 days it reaches 69%. The direction holds in **all 32** categories with enough volume to test and in **all 5** regions of Brazil.

> **Evidence:** RR 5.75 (95% CI 5.59–5.92) · Cliff's $\delta = -0.55$ (large) · Cramér's V = 0.37 (strong) · median category-level gap 45 pp

04 It is the broken promise, not the waiting, that customers punish

Lateness and slowness are correlated, so a raw correlation cannot separate them — and indeed the simple rank correlations slightly favour waiting time. A 2×2 that holds each constant settles it. Holding the wait short, missing the promise doubles the 1–2★ rate (8.1% → 15.9%); holding the wait long, it multiplies it five-fold (11.1% → 57.0%). But holding delivery on time, doubling the wait from 7 to 15 days moves the rate only 1.4× (8.1% → 11.1%). A long wait that was promised is largely tolerated. Since Olist already pads its estimates — the median order arrives **12 days early** — the estimated-delivery-date logic is itself an underused lever.

> **Evidence:** 2×2 on 96,470 delivered & reviewed orders, median wait 10 days · lateness multiplier 2.0×/5.1× vs wait multiplier 1.4×/3.6×

05 Geography and category are mostly delivery problems wearing a disguise

Supply is concentrated where demand is not: São Paulo hosts 59.7% of sellers but 42.0% of customers, so most parcels cross the country. Distance drives delivery time ($\rho = +0.54$) and freight ($\rho = +0.58$), and slower states rate lower ($\rho = -0.52$ across states). But restrict to on-time deliveries and the regional gap in 1–2★ rate collapses from 5.1% to 1.6%, and the category spread narrows from 1.32 to 0.89 stars. Distance correlates with the review score directly at only −0.06: it matters through what it does to timing, not in itself.

> **Evidence:** slowest state waits 3.6× the fastest (25.9 vs 7.2 days) · interstate penalty negligible as an effect size (Cliff's $\delta = -0.06$) · seller-revenue Gini 0.79

06 Payment behaviour explains spending, not satisfaction

Installments track basket size almost mechanically — median order value rises from R$60 at one installment to R$184 at ten or more — which is exactly what one expects in the Brazilian market. But payment type explains essentially none of the variance in review scores, and the apparent installment effect vanishes once order value is held constant: within every value quartile the gap is at most 1.2 percentage points and does not run consistently in one direction. Payment is a conversion lever, not a satisfaction lever.

> **Evidence:** installments vs order value $\rho = +0.37$ · payment type vs score $\eta^2 = 0.0001$ (negligible) · Cramér's V vs low review = 0.012

07 Sellers differ — and the difference is punctuality, not merchandise

After empirical-Bayes shrinkage (so tiny sellers with perfect scores stop topping the table) and after removing the score expected from each seller's category and destination mix, genuine performance differences remain: roughly half a star between the 10th and 90th percentile. The mechanism is fulfilment, not assortment — seller late rate correlates with shrunken score at $\rho = -0.46$, and 27 sellers miss the promised date significantly more often than chance explains. Critically, the bottom-decile sellers carry only 8.8% of revenue, so intervention is commercially cheap.

> **Evidence:** 795 sellers with ≥20 orders (87.7% of orders) · between-seller $\eta^2 = 0.030$ · residual spread −0.24 to +0.26 stars · 27 flagged by binomial test with BH-FDR

08 Half of all bad reviews are not about delivery at all — and the text says what they are

Decomposing every 1–2★ review into mutually exclusive experience buckets, 48.5% trace to a delivery problem and 51.5% come from orders that arrived on time and faster than average. Logistics cannot explain those. The written comments can: among bad reviews on fast, on-time orders, delivery language falls from 41% to 22% while product-quality and wrong-item language rises to 33%. There are two distinct problems on this platform, and they belong to two different owners.

> **Evidence:** 38,764 comments analysed (39% of reviews) · top negative terms *decepção* (disappointment), *devolução dinheiro* (money back), *descaso* (neglect), *enganosa* (misleading) · positive reviews praise the product first

SECTION 4

Supporting Visualizations

The eight charts that carry the argument, in the order the argument is built

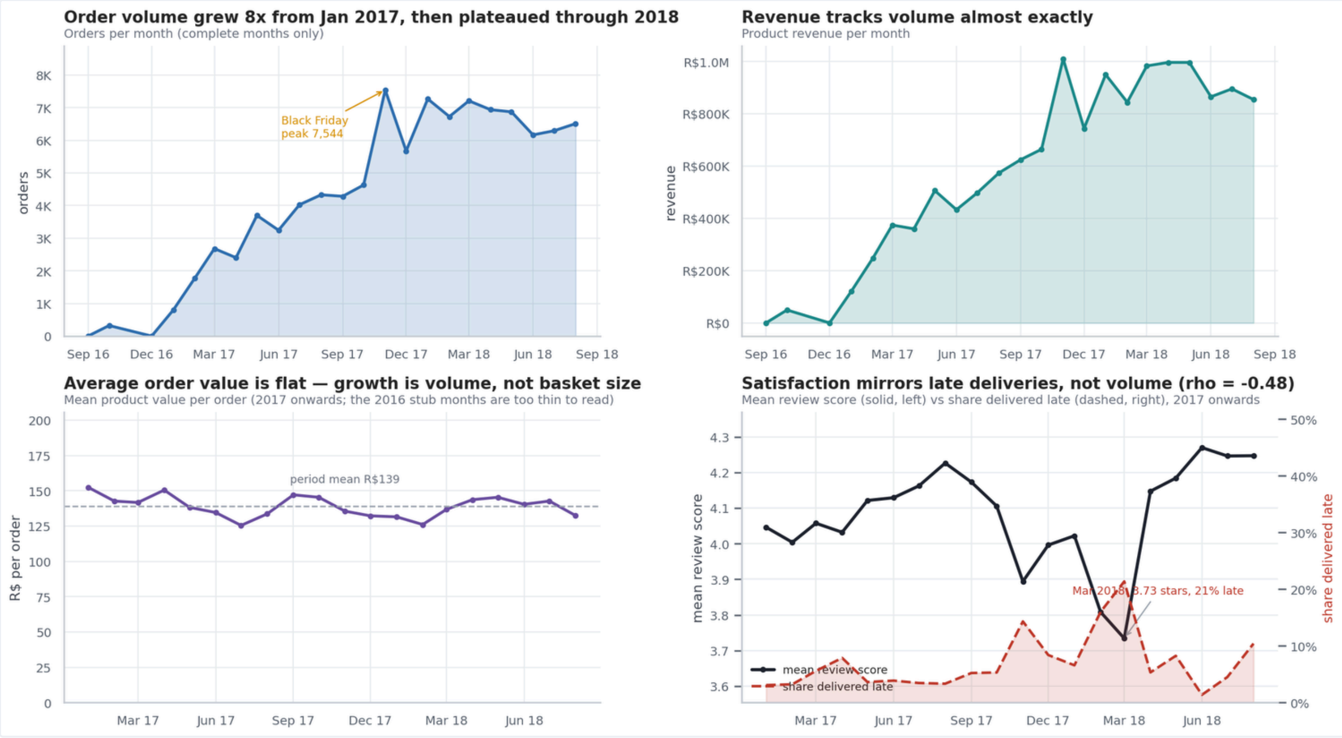

Figure 1 — Growth is a volume story; satisfaction is an execution story. Look at the bottom-right panel: the solid line (mean review score) and the dashed line (share delivered late) are near mirror images. Orders and revenue climb together while average order value stays flat, so growth came entirely from more orders. Satisfaction does not decline with that growth — it dips precisely when late deliveries spike (Nov 2017, Feb–Mar 2018) and recovers when they subside. *Population: complete months only, Sep 2016 – Aug 2018.*

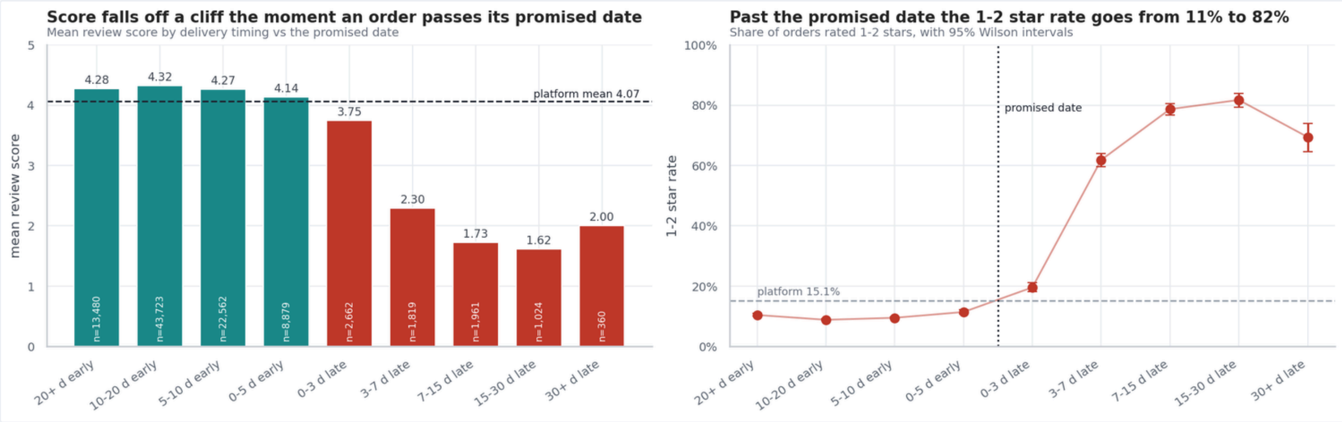

Figure 2 — Satisfaction falls off a cliff at the promised date. Look at the vertical divider: everything left of it arrived on or before the promise, everything right of it is late. Mean score drops from 4.14 to 3.75 in the first three days past the promise and keeps falling; the 1–2★ rate rises from roughly 10% to a peak of 82%. Error bars are 95% Wilson intervals — narrow enough that the ordering is not noise. The dose–response shape is what makes the causal reading plausible rather than merely correlational.

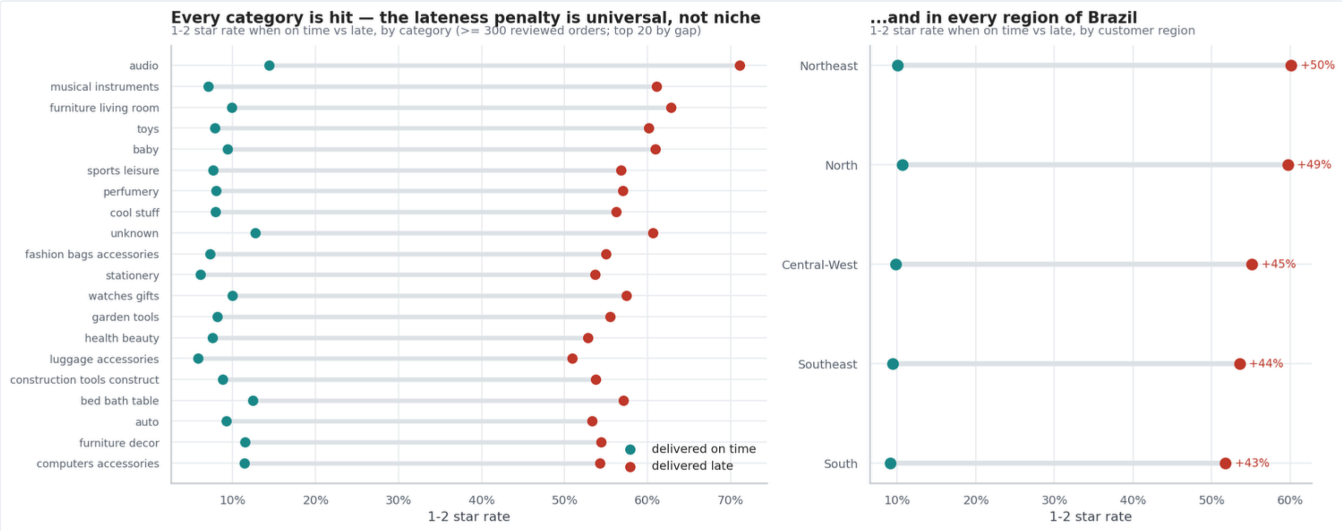

Figure 3 — The penalty is universal, so the fix must be structural. Each line joins the same group's 1–2★ rate when on time (teal) and when late (red); every line runs in the same direction. All 32 categories with sufficient volume and all 5 regions show the same penalty. This rules out a category-by-category or region-by-region remedy: the problem is the delivery process itself, not any particular product line or part of Brazil.

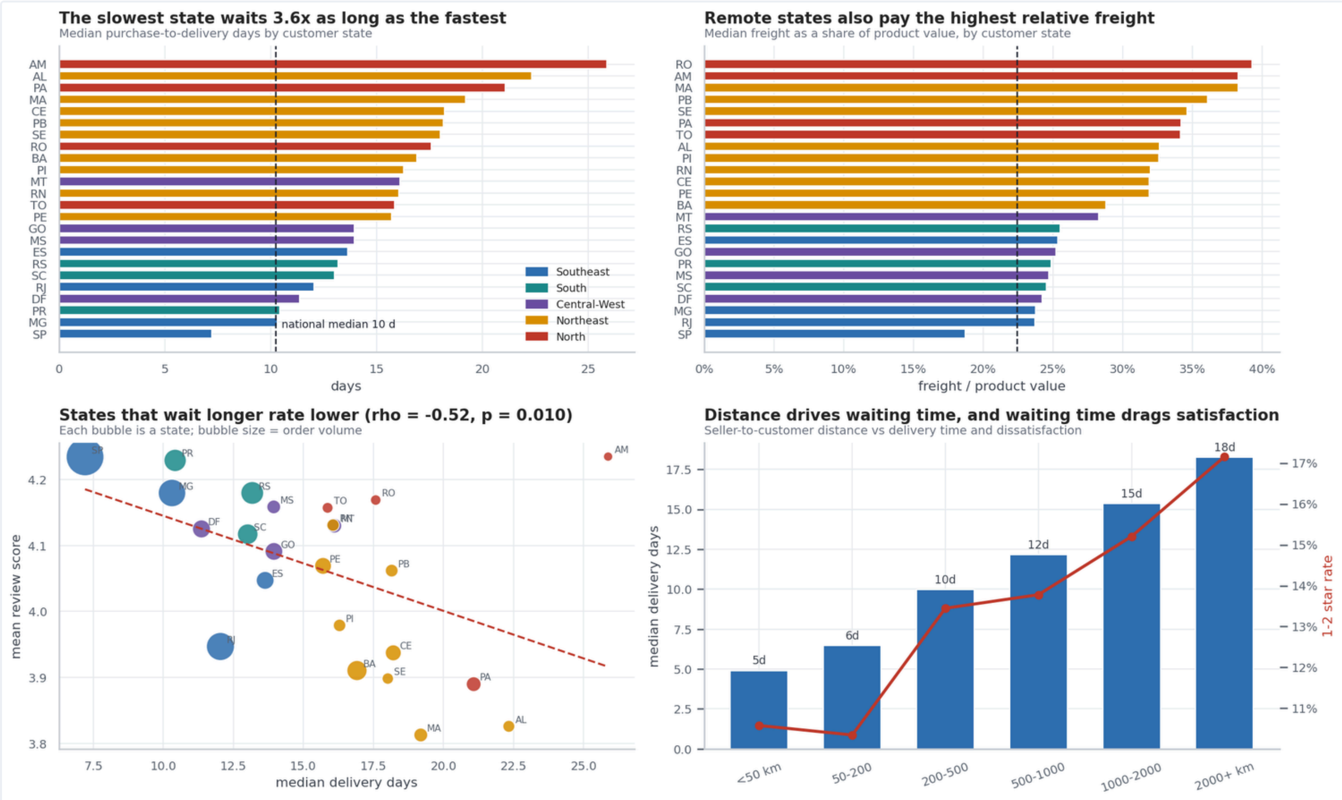

Figure 4 — Distance sets the difficulty, and difficulty shows up as time and freight. Compare the two top panels: the states at the top of the "slowest" list are largely the same states at the top of the "highest relative freight" list. Remote states are penalised twice — the slowest waits 3.6× as long as the fastest and pays freight worth over a third of the item value. The bottom-left scatter shows states that wait longer rate lower (ρ = −0.52); the bottom-right shows the mechanism, with delivery time and dissatisfaction both rising with seller–customer distance.

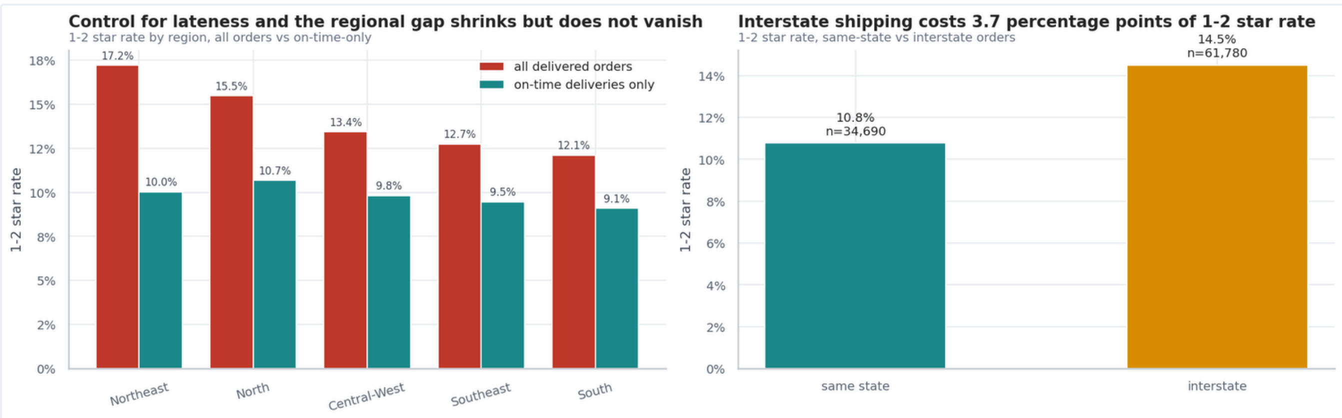

Figure 5 — Control for lateness and the regional gap nearly disappears. Compare the red bars (all orders) with the teal bars (on-time deliveries only) in the left panel. The spread in 1–2★ rate between best and worst region collapses from 5.1 points to 1.6. Remote customers are not intrinsically harder to please; they are disproportionately likely to receive a late parcel. This is the chart that turns "a regional satisfaction problem" into "a delivery problem that lands unevenly".

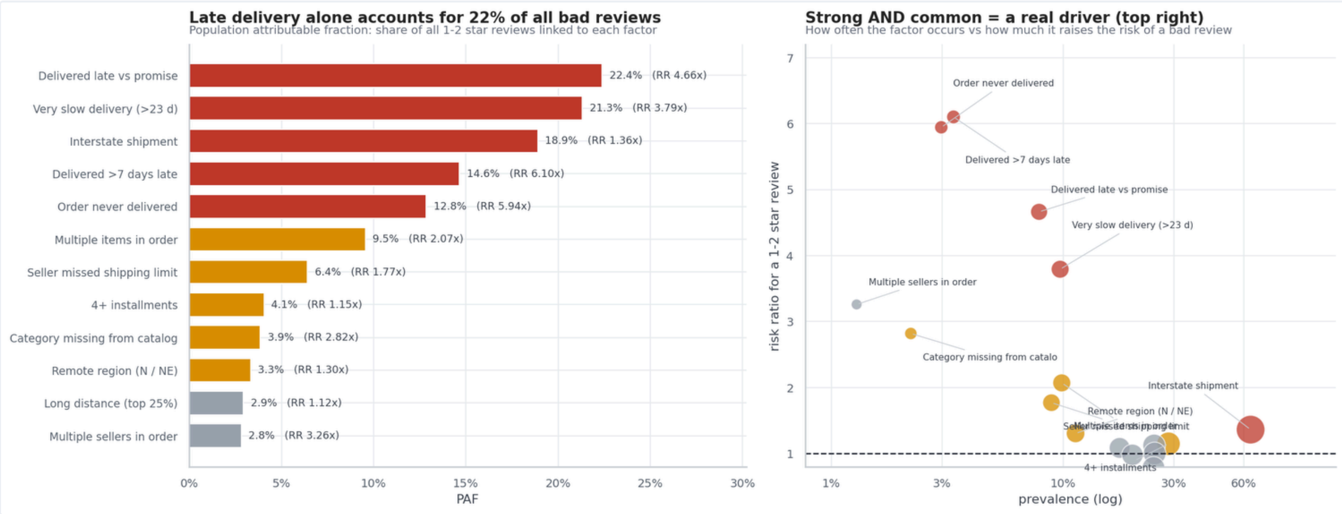

Figure 6 — Ranking drivers by damage done, not by strength alone. Read the left panel as "how much of the platform's dissatisfaction is attached to this factor"; the right panel separates common-but-mild factors from rare-but-severe ones. Delivery factors occupy the top of the ranking. Note the deliberate tension: being delivered more than 7 days late carries the highest risk ratio (6.1×) but a lower attributable share than plain lateness, because it is rarer. Interstate shipment ranks third on attributable share purely because it is so common — and it is exactly the kind of factor the deconfounding step then demotes.

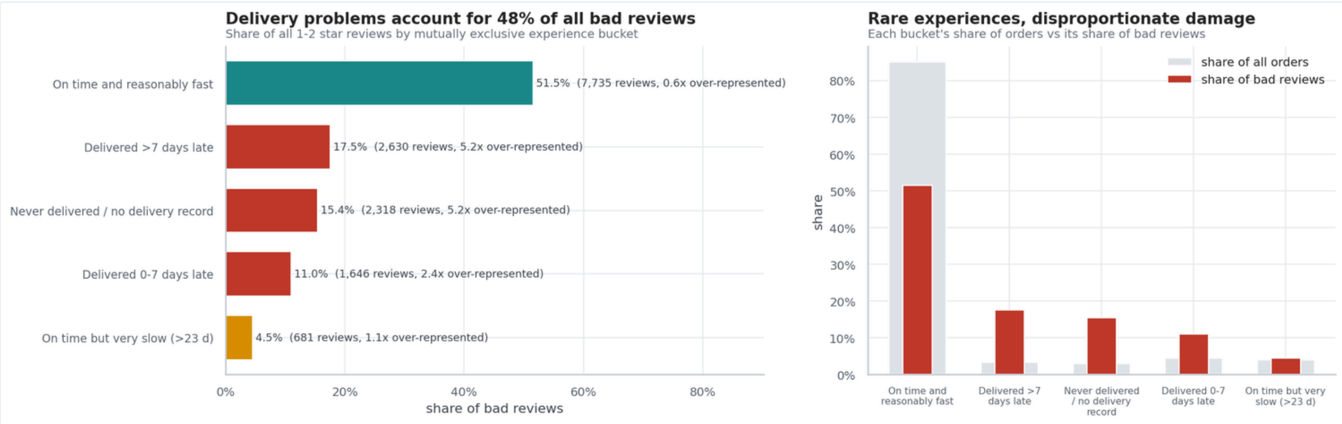

Figure 7 — Every bad review assigned to exactly one experience. The buckets are mutually exclusive and sum to 100%, so the teal bar is the part logistics cannot explain. Orders that never arrived produce 15.4% of all bad reviews from 3.0% of orders — 5.2× over-represented, the most damaging single experience on the platform. Severe lateness adds 17.5%. The 51.5% at the top is the residual that sends Section 5 looking at the review text.

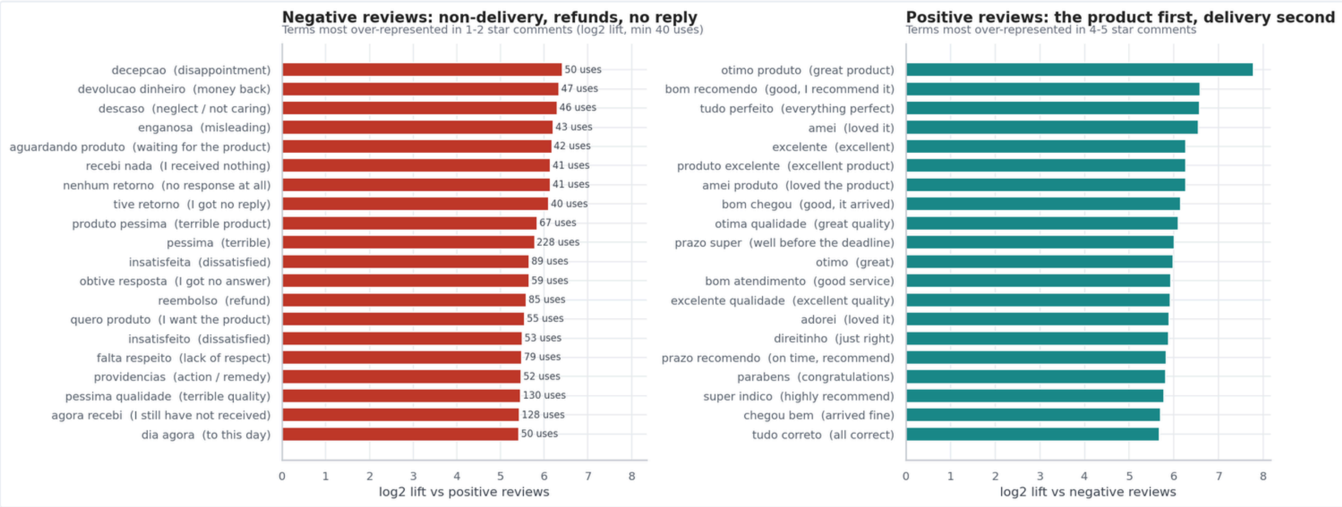

Figure 8 — An independent signal that agrees: what customers actually write. Terms are ranked by how much more often they appear in 1–2★ than in 4–5★ comments ($\log_2$ lift), with English glosses. Negative reviews cluster on non-delivery, refunds and silence — *não recebi* (I did not receive), *devolução dinheiro*, *descaso*, *nenhum retorno*. Positive reviews, notably, praise the *product* first (*ótimo produto*, *ótima qualidade*) with delivery praise secondary. Good delivery does not earn praise; it avoids punishment — the signature of a hygiene factor, and an argument for treating on-time delivery as a floor to defend rather than a feature to market.

SECTION 5

Business Findings

What is a primary driver, what is a passenger, and what it costs the business

The driver hierarchy after deconfounding

Twelve candidate factors were screened on the same population, then re-tested inside on-time deliveries only. That second pass is what separates a cause from a symptom: a factor whose risk ratio survives has an effect of its own; a factor whose effect collapses was only ever a proxy for lateness.

Table 3. Driver classification. PAF = share of all 1–2★ reviews attributable to the factor; RR = risk ratio for a 1–2★ review; "survives?" = does the effect persist among on-time deliveries.

Factor	Prevalence	1–2★ rate	RR	PAF	Survives?	Verdict
Delivered late vs promise	7.9%	54.6%	4.66	22.4%	—	PRIMARY
Very slow delivery (>23 days)	9.7%	45.0%	3.79	21.3%	Yes — 1.88×	PRIMARY
Delivered >7 days late	3.4%	78.6%	6.10	14.6%	—	PRIMARY
Order never delivered	3.0%	78.2%	5.94	12.8%	—	PRIMARY
Seller missed shipping limit	8.9%	25.0%	1.77	6.4%	Yes — 1.53×	SECONDARY
Category missing from catalogue	2.2%	40.9%	2.82	3.9%	Yes	SECONDARY
Multiple items in order	9.9%	28.3%	2.07	9.5%	Partly	SECONDARY
Multiple sellers in order	1.3%	47.8%	3.26	2.8%	Partly	SECONDARY
Interstate shipment	64.3%	16.7%	1.36	18.9%	No	CONFOUNDED
Remote region (N / NE)	11.3%	19.0%	1.30	3.3%	No	CONFOUNDED
Long distance (top 25%)	24.7%	16.4%	1.12	2.9%	No	CONFOUNDED
High freight ratio (top 25%)	24.8%	15.1%	1.00	0.1%	No	NOT A DRIVER
4+ installments	28.5%	16.6%	1.15	4.1%	No	NOT A DRIVER
Paid by boleto	19.9%	14.9%	0.98	—	No	NOT A DRIVER

The interstate row is the clearest illustration of why the second pass is necessary. On raw attributable share it ranks third on the entire platform — a finding that would justify a national logistics programme. Once lateness is held constant its effect disappears. It was never a driver; it was a marker for the shipments most likely to run late.

What the dissatisfaction actually costs

Bad reviews are not the only cost. Retention is measurably attached to the same operational failure. Only 3.12% of customers ever place a second order — very low for e-commerce even allowing for the two-year window — and among customers whose first order came early enough to have a fair chance to return, a late first delivery is associated with a 22% lower chance of ever coming back.

Table 4. Business consequences quantified across the 25-month period

Consequence	Magnitude	Reading
Bad reviews from delivery failure	7,275 of 15,010	48.5% of all 1–2★ reviews, from a small minority of shipments.
Over-representation of late orders	4.2×	8.1% of deliveries produce 33.7% of bad reviews.
Never-delivered orders	5.2×	3.0% of orders produce 15.4% of bad reviews — the most damaging single experience.
Retention penalty after a late first order	−22%	Return rate 3.78% → 2.94% (RR 0.78, 95% CI 0.67–0.90).
Value of a repeat customer	2.1×	Median lifetime revenue R$180 vs R$86 for one-time buyers.
Revenue exposed to worst-decile sellers	8.8%	Low enough that SLA enforcement carries little commercial risk.
Operational exceptions requiring action	3,314 orders	3.3% of orders trip at least one anomaly rule; 642 exceed the extreme-delay fence (max 189 days late).

AN HONEST NOTE ON THE RETENTION NUMBER

The retention effect is statistically solid (the confidence interval excludes 1.0) but small in absolute terms — 0.84 percentage points — because the base rate is itself so low. Worth noting too: the bad *score* predicts return behaviour more weakly than the late *delivery* does (3.46% vs 2.94%). It is the operational failure that shows up in retention, not the act of leaving a bad review.

The second problem: product and listing accuracy

The 51.5% of bad reviews attached to fast, on-time orders is too large to leave unexplained. The written comments identify it: among those orders, product-quality and wrong-item language appears in 33% of negative comments while delivery language falls to 22%. The catalogue data agrees — products whose category is missing entirely average **3.14 stars** against a platform mean of 4.07, the single worst group tested, and the effect survives the deconfounding pass. Incomplete listings are associated with materially worse outcomes, which is a merchandising and catalogue-governance problem rather than a logistics one.

SECTION 6

Actionable Recommendations

Six priorities, sized by the dissatisfaction each would remove

The scenarios below are **association-based upper bounds**, not causal forecasts: each asks what the platform's 1–2★ rate would look like if the exposed orders had behaved like comparable unexposed ones. They are deliberately conservative — "eliminate lateness" removes 23.5% of bad reviews rather than the 33.7% that late orders carry, because on-time orders also generate bad reviews at a baseline rate of 9.5%.

Table 5. Opportunity sizing over the 25-month period, ranked by recoverable dissatisfaction

Intervention	Orders affected	1–2★ reviews avoided	% of all bad reviews	Platform 1–2★ rate
Eliminate all late deliveries (theoretical ceiling)	7,826	3,533	23.5%	15.1% → 11.5%
Fix only the >7-day-late tail	3,345	2,312	15.4%	15.1% → 12.8%
Halve the late rate (realistic 12-month target)	3,913	1,766	11.8%	15.1% → 13.3%
Remediate the worst-decile sellers	7,488	1,194	8.0%	15.1% → 13.9%

Note the efficiency argument hiding in the second row: fixing only the severe tail touches **less than half** as many orders as eliminating lateness altogether, yet recovers nearly two-thirds as much dissatisfaction. That is where the first engineering effort should go.

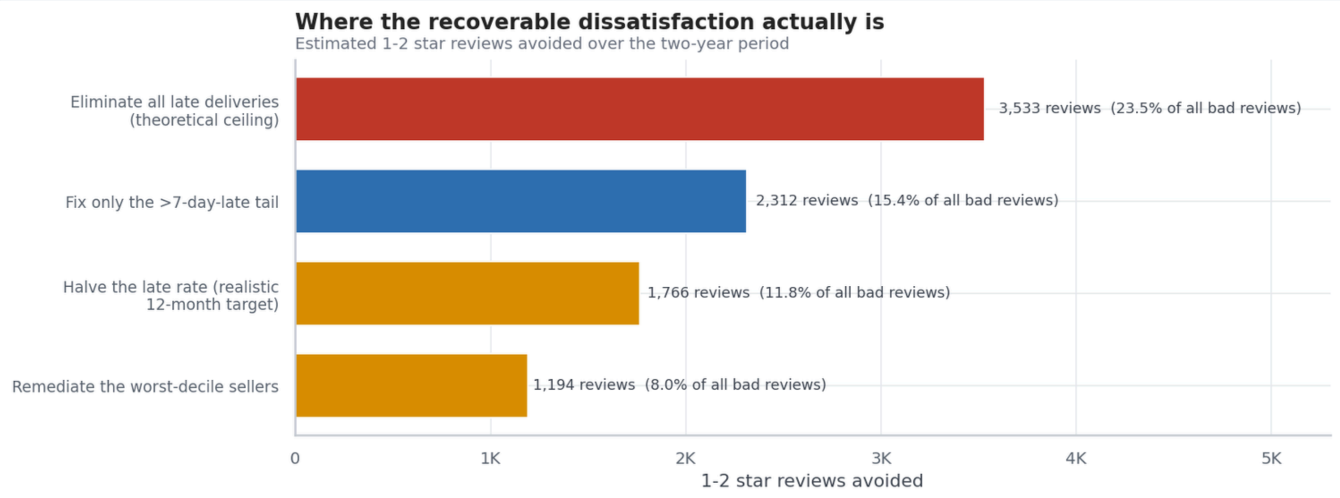

Figure 9 — Where the recoverable dissatisfaction sits. Bar length is the number of 1–2★ reviews each intervention would avoid over the period. The severe-lateness tail (blue) is the efficiency play: fewer orders touched, most damage removed per order fixed. Seller remediation (red) is the cheapest to execute given how little revenue the worst decile carries.

P1 **Manage the promise, not just the parcel**

Make "delivered by the promised date" the primary operational KPI, reported by seller, lane and category — replacing average delivery time, which the data shows customers care about far less. Where the network cannot be made faster, **widen the estimate**: an honest longer promise outperforms an optimistic broken one, because customers punish the miss (2.0–5.1×) far more than the wait (1.4×). Estimates are already padded by a median of 12 days, so this is a calibration exercise on an existing system, not new infrastructure.

EXPECTED EFFECT	OWNER	EFFORT	TRACK
Up to 3,533 fewer 1–2★ reviews (23.5% of all bad reviews)	**Logistics & Ops**	**Low — recalibration, not build**	**On-time-vs-promise rate by lane**

P2 Build an exception queue for non-delivery and the >7-day tail

Orders that never arrive (5.2× over-represented in bad reviews) and orders more than a week late (78.6% rated 1–2★) are the two most damaging experiences on the platform, together carrying roughly a third of all bad reviews from a very small share of volume. Trigger proactive contact — and where possible compensation — the moment an order passes its promised date, *before* the satisfaction survey is sent. Recall that Olist itself sends that survey on the promised date, which means the current process invites a review from customers it has already failed.

EXPECTED EFFECT	OWNER	EFFORT	TRACK
2,312 fewer 1–2★ reviews (15.4%), from 3,345 orders	**Customer Experience**	**Medium — workflow + alerting**	**Exceptions contacted before survey**

P3 Enforce seller fulfilment SLAs, and auto-pad the offenders

Missing the contractual shipping limit raises bad-review risk 1.53× *even when the parcel still arrives on time*, which places part of the problem upstream of the carrier. Publish each seller's on-time-handover rate, escalate the 27 sellers that fail significantly more often than chance, and automatically widen delivery estimates for sellers with a persistently poor record. The bottom decile carries just 8.8% of revenue, so enforcement — up to delisting — costs little.

EXPECTED EFFECT	OWNER	EFFORT	TRACK
1,194 fewer 1–2★ reviews (8.0%)	**Seller Operations**	**Low — data already exists**	**Shrunken seller score residual**

P4 Attack distance on the worst lanes — the slow, structural lever

The São Paulo supply concentration (59.7% of sellers, 42.0% of customers) is the structural cause behind long transit times and high freight in the North and Northeast. Recruiting sellers or placing fulfilment capacity closer to those customers attacks distance, delivery time and freight burden at once. This is the only recommendation that changes the underlying constraint rather than managing it — and correspondingly the slowest and most capital-intensive. Sequence it after P1–P3, not instead of them.

EXPECTED EFFECT	OWNER	EFFORT	TRACK
Structural; reduces the population exposed to P1–P3 risks	**Network Strategy**	**High — capital & recruitment**	**Median seller–customer distance by region**

P5 Close the catalogue-accuracy gap behind the other half of bad reviews

51.5% of bad reviews come from well-delivered orders, and the customers' own words point at damage, wrong items and quality. Start where the signal is strongest and cheapest to act on: the 610 products with no category at all average 3.14 stars against a 4.07 platform mean. Enforce catalogue completeness at listing time, then run a listing-accuracy and packaging review on the categories where wrong-item and damage language concentrates.

EXPECTED EFFECT	OWNER	EFFORT	TRACK
Addresses the 51.5% logistics cannot reach	**Catalogue & Merchandising**	**Medium — validation rules + review**	**1–2★ rate on on-time, fast orders**

P6 **Treat retention as a delivery outcome, and measure it that way**

With a repeat rate near 3%, every retained customer is disproportionately valuable — repeat buyers are worth 2.1× a one-time buyer in median lifetime revenue. Since a late first delivery is associated with a 22% lower chance of a second order, first-order delivery quality compounds: it protects the rating *and* the customer. Make "return rate split by first-order delivery outcome" a standing metric so the compounding effect of P1–P3 becomes visible to the business.

EXPECTED EFFECT	OWNER	EFFORT	TRACK
Makes the compounding value of P1–P3 measurable	**Growth / Analytics**	**Low — reporting only**	**Repeat rate by first-order outcome**

APPENDIX A

Method, limitations & reproducibility

What this analysis cannot claim, and how to re-run it

Limitations

- **Observational, not experimental.** Every relationship reported is an association. The delivery finding carries a large effect, a clean dose–response, stability across every stratification tested and independent corroboration from review text — which is as strong as observational evidence gets — but the dataset contains no intervention, so causation is inferred, not proven.
- **Written comments are a self-selected sample.** Only 39% of reviews carry text, and unhappy customers write far more often than happy ones. Text-derived percentages are directional; the keyword method has no negation or sarcasm handling.
- **The window is truncated.** Orders placed near October 2018 had not been delivered or reviewed at export. All trend work excludes September–October 2018; cross-sectional work includes them and is only marginally affected.
- **No cost data.** Freight is what the customer paid, not what Olist paid. Recommendations are sized in review-score impact, not in margin, and P4 in particular would need a cost case before commitment.
- **Repeat behaviour may be understated** if customers re-register rather than reuse an account; `customer_unique_id` is a derived key, not an authoritative one.
- **Seller attribution** assigns each order's review to its most expensive line's seller, so multi-seller orders (1.3%) credit one seller with the whole outcome.

Data quality observed

All nine tables load at their stated row counts and every foreign key resolves — there are no orphan items, payments or reviews. The gaps run the other way: 775 orders have no item line, and 2,965 lack a delivery timestamp, consistent with orders that never completed fulfilment. Anomaly screening flagged 3,314 orders (3.3%), of which the operationally important group is the 2,963 orders reviewed despite never being marked delivered. Only 23 orders show a physically impossible timestamp sequence, and 148 orders are price outliers when judged within their own category — a catalogue that is, on the whole, sane.

Reproducibility

The companion notebook, `olist_marketplace_analysis.ipynb`, runs top to bottom without errors (64 cells: 45 code, 19 markdown) and regenerates every number and chart in this report. It was validated by full execution against the source data; all 45 code cells complete cleanly in roughly five minutes. Random operations are seeded. Headline figures are collected into a registry during the run and used to generate the notebook's own executive summary, so the narrative cannot drift from the computed values. The notebook also exports nine CSV artefacts, including the order-level master table, per-state, per-category and per-seller KPI tables, the ranked driver table behind Table 3, and the flagged-anomaly lists behind the operational recommendations.

Source data: Brazilian E-Commerce Public Dataset by Olist, published on Kaggle under a CC BY-NC-SA licence. Approximately 100,000 orders placed between September 2016 and October 2018, anonymised by Olist before release; customer and seller identities are hashed and company names in review text were replaced with fictional names. Verify current licence terms before any commercial or redistribution use.
Analysis populations: 99,441 orders total · 96,470 delivered with a verified delivery timestamp · 96,470 delivered and reviewed · 38,764 reviews with usable written text.